MILITARY SPECIFICATION
CARTRIDGE, CALIBER .30, BALL, M2 OVERHEAD FIRE APPLICATION

1. SCOPE

1.1 This specification covers Cartridge, Caliber .30, Ball M2 Overhead Fire Application, for use in Caliber .30 weapons.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition.

DRAWINGS

Munitions Command

C10542449	- Cartridge, Caliber .30, Ball, M2, Overhead Fire Application
B8595559	- Bullet
C7643674	- Classification of Cartridge Case Defects
F10542451	- Packing and Marking, Cartridge, Caliber .30, Ball, M2 Overhead Fire; Linked; Box, Ammunition, M19A1; Box, Wirebound.
IEL 10542449	- List of Inspection Equipment for Cartridge Caliber .30, Ball, M2, Overhead Fire Application.

FSC 1305

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PUBLICATIONS

Munitions Command

AMCR 715-505, Vol. III	-	Ammunition Ballistic Acceptance Test Methods and Procedures for 7.62mm Cartridges.
TECP 700-700, Vol. III	-	Manual of Test Methods for Small Arms Ammunition.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with Drawing D10542449, referenced specifications and the following:

3.1.1 Bullet.- Prior to assembly into cartridges, all bullets shall be examined for conformance with Drawing B8595559 and shall be free from dents, scratches, splits, scaly metal, defective slug, defective base and point, and defective or missing cannelure.

3.2 Bullet extraction.- The force necessary to extract the bullet from the cartridge case shall be not less than 45 pounds.

3.3 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.4 Accuracy.- The average of the mean radii of all targets of the sample cartridges fired at 600 yards shall not exceed 5.0 inches. The extreme spread of each target of the sample cartridges shall not exceed 25 inches.

3.5 Action time.- The action time (overall primer ignition, propellant burning and bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

3.6 Velocity.- The average velocity of the sample cartridges conditioned at 68° to 72° Fahrenheit (F), shall be 2740 feet per second (ft/sec) plus or minus 30 ft/sec at 78 feet from the muzzle of the weapon. The standard deviation of the velocities shall not exceed 20 ft/sec.

3.7 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 30 seconds.

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3.8 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at 68° to 72°F, shall not exceed 50,000 psi.

3.9 Function and casualty.- The cartridge shall function without casualty and neither the neck of the cartridge case nor the chamber of the weapon shall show any evidence of deformation (peening) after firing.

3.10 Stripping.- The jacket of the bullet, or any part thereof, shall not strip from the slug when the cartridge is fired.

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects.

3.11.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 2500 cartridges and 500 bullets. The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the supplier's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- Examination shall be performed as indicated below:

4.3.2.1 Bullet examination.- Prior to assembly into a cartridge, each bullet shall be weighed and visually examined for:

a. Dents
b. Scratches
c. Split jacket
d. Scaly metal
e. Partial or missing slug
f. Improper basing
g. Upset or blunt points
h. Defective or missing cannelure
i. Weight

The presence of any one of these defects shall be cause for rejection of the bullet. The bullet weighings shall be accomplished on equipment capable of weighing over a range of 140 to 155 grains with an accuracy of 0.5 grains.

4.3.2.1.1 The bullets extracted from the cartridge in the bullet extractor test shall be examined for:

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4.3.2.1.1 (Cont'd)

a. Split jacket
b. Scaly metal
c. Partial or missing slug
d. Improper basing.

The presence of any of these defects shall be cause for rejection of the cartridge lot.

4.3.2.2 Cartridge examination .- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of Standard MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.10 percent and the AQL for the minor class shall be 1.00 percent. All non-conforming cartridges shall be rejected.

4.3.2.3 Classification of defects.- The classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection Visual 1/	Critical	Major	Minor	Major or Minor
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case, any location	X			
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Case mouth not crimped in cannelure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent		X		
23	Scratch		X		
24	Split bullet jacket		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal (Bullet)		X		
28	Upset (crooked) point		X		
30	Blunt point		X		
31	Defective cannelure		X		
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	
	Gaging				
39	Total length		X		
40	Cartridges profile failure (requiring more than 20 lbs, dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum		X		
46	Depth of primer		X		
47	Weight, min.	X			

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1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 38. In event of conflict between Table I of this specification and MIL-STD-636 as to defect classification, the classification specified in Table I shall apply.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representative of the entire lot. The cartridges selected shall be thoroughly mixed before being divided into samples for the various tests.

Table II

Test	Number of Cartridges	Requirement Paragraph
Bullet extraction 1/	25	3.2
Residual stress (Mercurous nitrate) 1/	50	3.3
Accuracy 2/	90	3.4
Action time 1/	50	3.5
Velocity 6/	20	3.6
Waterproof 4/	20	3.7
Chamber pressure 6/	20	3.8
Stripping 5/		3.10
Function and casualty 3/		3.9
Gun, Machine, Cal. .30, M37 (Tank) or Alternate Gun, Machine Cal. .30, Browning, M1919A4	300	
Gun, Machine, Cal. .30 M1919A6	300	
Rifle, U.S. Cal. .30 M1	408	

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the initial test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any defective cartridges are found in the second sample, the lot shall be rejected.

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4.3.3.1 (Cont'd)

2/ Failure of two or more targets to comply with the extreme spread requirement shall be cause for rejection of the lot. If one target fails to meet the extreme spread requirement in the first test, a second sample consisting of double the number of targets in the first sample shall be tested. Failure of one or more targets of the second sample to comply with the extreme spread requirement shall be cause for rejection of the lot. Failure of the cartridges to comply with the average mean radius requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the average mean radius requirement shall be cause for rejection of the lot.

3/ The lot shall be rejected when function and casualty defects plus all other firing defects exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the initial test exceeds the acceptance number for the first sample (Table III) but is equal to or less than the acceptance number for the cumulative sample (Table III), a second sample, consisting of double the quantities specified under function and casualty test, shall be fired in all the service weapons specified therefor. This procedure shall apply regardless of the weapon or weapons in which the firing defects occurred during the initial test. If the total number of defects in the combined first and second sample exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If, in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance number for the cumulative sample, the lot shall be rejected.

4/ Failure of nine or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than four, but less than nine cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, nine or more cartridges fail to comply with the applicable requirement.

5/ Determination of compliance with the bullet stripping requirement shall be made during the function and casualty test.

6/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot, subject to testing a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

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4.3.3.2 (Cont'd)

TABLE III

Defects	Acceptance Number: First Sample	Acceptance Number: Cumulative (First & Second Sample)
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup		
(1) Machine guns	12	28
(2) Rifles	2	4
b. Escape of gas through primer cup other than 3a.	1	2
c. Escape of gas around primer cup		
(1) More than 50% of periphery	10	21
(2) Less than 50% of periphery	24	53
d. Blown primer or primer falls out pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	4	7
4. Case Casualties		
a. Longitudinal split 2/		
(1) Neck and shoulder (I & S)	24	53
(2) Body (J)	4	7
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 2/		
(1) Partial, shoulder or body (J & S)	1	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1
5. Failure to extract	0	1
6. Weapon stoppage 3/	0	1
7. Deformation (Peening) 1/	0	-
8. Bullet stripping 1/	0	-

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4.3.3.2 (Cont'd)

1/ No second sample permitted. Lot shall be rejected.

2/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

3/ All stoppages attributable to the ammunition, with the exception of misfire or complete rupture, observed in all tests shall be included.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100 percent examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of any incorrect type shall be classed as a critical defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-10542449. Simulated assessment of reference cartridges shall be made in accordance with TECP 700-700, Vol. III.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute. The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.2 Residual stress (mercurous nitrate).- Tests shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.3 Accuracy.- The test shall be conducted in accordance with TECP 700-700, Vol. III, utilizing the longest range available up to 600 yards. Ranges shorter than 200 yards shall not be used. When accuracy testing is conducted on ranges shorter than 600 yards, the average of the mean radii of the targets and the extreme spread of each target shall be multiplied by the applicable conversion factor given below:

Range (yards)	200	300	400	500
Factor	3.40	2.23	1.68	1.35

4.4.4 Action time.- The test shall be conducted in accordance with AMCR 715-505, Volume 3.

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4.4.5 Velocity.- The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.6 Waterproof.- The test shall be conducted in accordance with AMCR 715-505, Vol. 3.

4.4.7 Chamber pressure.- The cartridges shall be drilled, using the specified equipment and the drilled hole shall be covered with a half inch square of tape. The test shall be conducted in accordance with TECP 700-700, Vol. III.

4.4.8 Function and casualty.- In these firings the weapon shall be at room temperature at the beginning of the test and the machine guns shall be cooled between bursts. The number of cartridges to be fired shall be as specified in Table II. The test shall be conducted in accordance with TECP 700-700, Vol. III and as indicated below:

a. Gun, Machine, Caliber .30, M37 (Tank) or alternate Gun, Machine, Caliber .30, Browning, M1919A4 - Fire in bursts of 100 cartridges.

b. Gun, Machine, Caliber .30, Browning, M1919A6 - Fire in bursts of 100 cartridges.

c. Rifle, U.S. Caliber .30, M1 - Fire 8 cartridges (one clip) rapid fire, with intervals of not more than 1/2 minute between clips. The weapon shall be cooled after firing the first 48 cartridges (6 clips). Two rifles shall be used and 104 cartridges shall be fired in each rifle.

d. Rifle, U.S. Caliber .30 M1 - One hundred (100) cartridges in each of two rifles shall be fired single shot. After the cartridge has been chambered and the rifle bolt locked in the firing position, the rifle muzzle shall be elevated to at least a 45° angle above horizontal. The weapon shall then be lowered slowly to a position not below horizontal in a manner that assures the propellant remaining to the rear of the cartridge case at the instant of firing.

4.4.8.1 Deformation (peening).- The expended cartridge cases and the chambers of the weapons shall be examined after the completion of the firing specified in (a), (b) and (c) above. After each shot fired in (d) above the expended cartridge case shall be visually inspected for any evidence of deformation, the chambers of the weapons used shall be examined at the conclusion of the test. Any evidence of peening deformation of the weapon chamber or any evidence of a projection or localized expansion in the neck of the cartridge case shall be considered unacceptable.

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4.4.9 Stripping.- The test shall be conducted in accordance with TECP 700-700, Vol. III simultaneously with the function and casualty test.

4.4.10 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect, listed in Table III prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table III. Any abnormal condition not described herein shall be recorded and evaluated by the procuring agency prior to acceptance of the ammunition lot.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A (Worldwide shipment).- The cartridge shall be packed in accordance with Drawing F10542451.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawings cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provisions for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Requirements for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

CUSTODIAN:

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PREPARING ACTIVITY:

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